

Chp 6.5: Inverse Trigonometric Functions and Their Graphs**I. The Inverse Sine Function**

The **inverse sine function** is the function $\sin^{-1}$ with domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$ defined by $\sin^{-1} x = y \Leftrightarrow \sin y = x$. The inverse sine function is also called arcsine, denoted by arcsin.

$y = \sin^{-1} x$ is the number in the interval $[-\pi/2, \pi/2]$ whose sine is x .

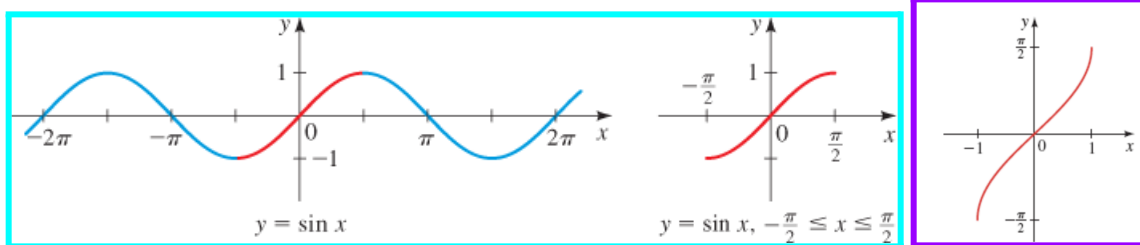
The inverse sine function has these **cancellation properties**:

$$\sin(\sin^{-1} x) = x \text{ for } -1 \leq x \leq 1$$

$$\sin^{-1}(\sin x) = x \text{ for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

Graph of $y = \sin^{-1} x$

**II. The Inverse Cosine Function**

The **inverse cosine function** is the function $\cos^{-1}$ with domain $[-1, 1]$ and range $[0, \pi]$ defined by $\cos^{-1} x = y \Leftrightarrow \cos y = x$. The inverse cosine function is also called arccosine, denoted by arccos.

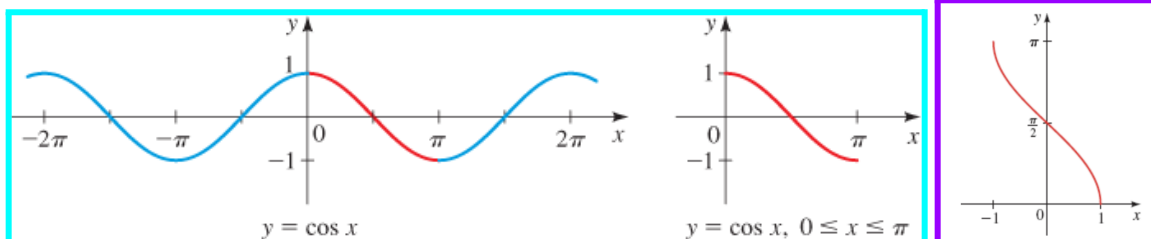
$y = \cos^{-1} x$ is the number in the interval $[0, \pi]$ whose cosine is x .

The inverse cosine function has these **cancellation properties**:

$$\cos(\cos^{-1} x) = x \text{ for } -1 \leq x \leq 1$$

$$\cos^{-1}(\cos x) = x \text{ for } 0 \leq x \leq \pi$$

Graph of $y = \cos^{-1} x$

**III. The Inverse Tangent Function**

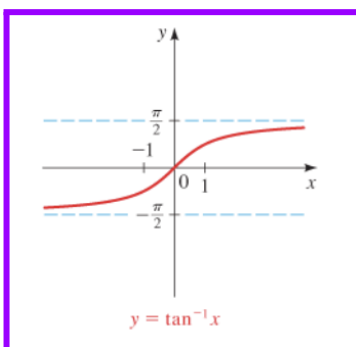
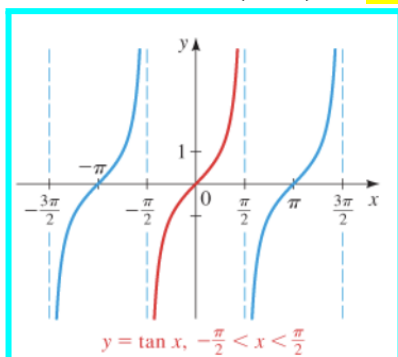
The **inverse tangent function** is the function $\tan^{-1}$ with domain $\mathbb{R}$ and range $(-\pi/2, \pi/2)$ defined by $\tan^{-1} x = y \Leftrightarrow \tan y = x$. The inverse tangent function is also called arctangent, denoted by arctan.

$y = \tan^{-1} x$ is the number in the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ whose tangent is x .

The inverse tangent function has these **cancellation properties**:

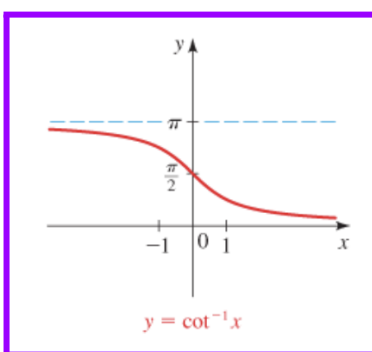
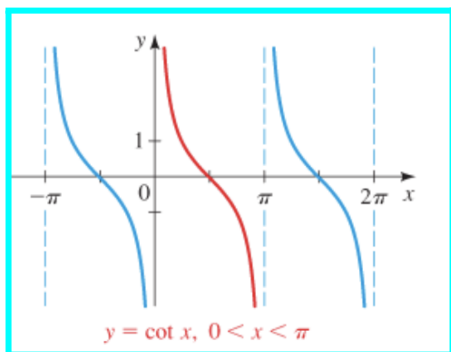
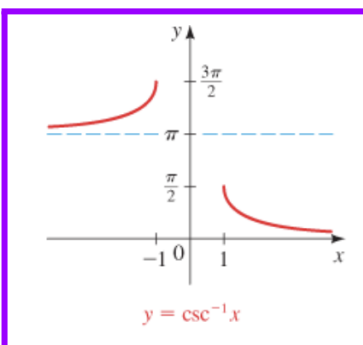
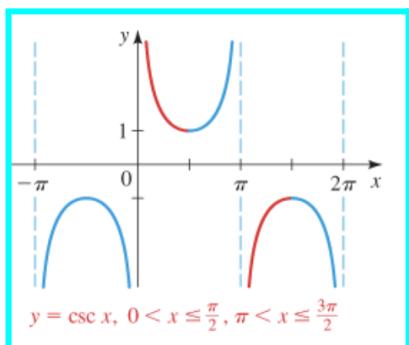
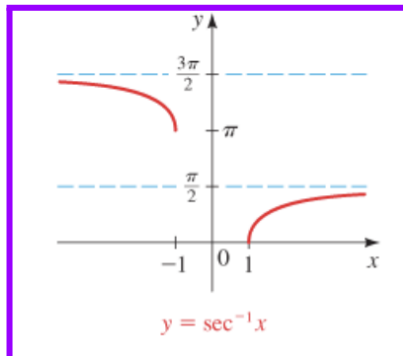
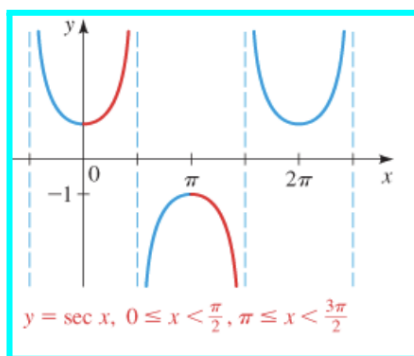
$$\tan(\tan^{-1}x) = x \text{ for } x \in \mathbb{R}$$

$$\tan^{-1}(\tan x) = x \text{ for } -\frac{\pi}{2} < x < \frac{\pi}{2}$$



The Inverse Secant, Cosecant, and Cotangent Functions

To define the inverse functions of the secant, cosecant, and cotangent functions, we restrict the domain of each function to a set on which it is one-to-one and on which it attains all its values.



Example 1: Find the exact value of each expression, if it is defined.

a. $\tan^{-1} 1$

b. $\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$

c. $\sin^{-1}(-1)$

Example 2: Use a calculator to find an approximate value of each expression correct to five decimal places, if it is defined.

a. $\sin^{-1}(0.13844)$

b. $\cos^{-1}\left(\frac{-3}{7}\right)$

Example 3: Find the exact value of the expression, if it is defined.

a. $\sin\left(\sin^{-1}\left(\frac{1}{4}\right)\right)$

b. $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$

c. $\tan^{-1}\left(\tan\frac{2\pi}{3}\right)$

d. $\sin(\sin^{-1} 5)$

e. $\sin(\sec^{-1}(-2))$

f. $\csc(\tan^{-1}(-\sqrt{3}))$

g. $\tan\left(\cos^{-1}\left(\frac{-\sqrt{2}}{2}\right)\right)$

Example 4: Rewrite the expression as an algebraic expression in x .

a. $\cos(\sin^{-1} x)$

b. $\sec(\tan^{-1} x)$

Example 5: Find functions f and g such that $F = f \circ g$.

a. $F(x) = e^{\arcsin x}$

b. $F(x) = \left(\tan^{-1} x\right)^2$